Benchmark report

Table 1. Final model vs open baselines and frontier models (fresh eval_id v2, 300 cases)

Table 1a. Decision quality

Model	Params	Cases	Decision acc %	Their acc − ours (pts) [95% CI, paired]	Screening macro-F1 %	Risk exact %	Risk ±1 %	Actions recall %	Actions precision %	Unsupported claims % (LLM judge)	Schema valid %
Ours: KYC specialist (B2), Qwen3.5-4B	4B	300	96.3	–	99.6	95.3	100.0	93.6	97.6	13.7	100.0
Qwen3.5-0.8B (best prompt)	0.8B	300	44.3	-52.0 [-57.7, -46.3]	25.2	26.3	83.7	4.7	32.0	60.8	100.0
Qwen3.5-2B (best prompt)	2B	300	39.7	-56.7 [-62.3, -51.0]	42.9	37.0	88.3	30.2	17.8	45.3	100.0
Qwen3.5-4B (best prompt)	4B	300	56.0	-40.3 [-46.3, -34.0]	34.6	50.3	98.3	34.6	17.3	52.0	100.0
GPT-6 Astra (best prompt)	undisclosed	300	99.3	+3.0 [+0.7, +5.3]	100.0	99.3	100.0	98.8	99.7	7.4	100.0
Claude Opus 5.5 (best prompt)	undisclosed	300	97.7	+1.3 [-1.3, +4.0]	100.0	97.3	100.0	97.4	96.3	8.6	100.0
Gemini 4 Argon (best prompt)	undisclosed	300	95.0	-1.3 [-4.7, +2.0]	100.0	95.7	100.0	92.2	96.1	8.8	100.0

Table 1b. Identity fields, adversarial cases, latency and cost

Model	Verified %	Name %	Date of birth %	Document no. %	Expiry %	Address %	Escalation on injection % (escalated / should escalate)	Escalation on ambiguous evidence % (escalated / should escalate)	p50 latency	p95 latency	$ / 1k cases
Ours: KYC specialist (B2), Qwen3.5-4B	98.3	99.3	99.7	99.3	100.0	98.3	86.2 (25/29)	100.0 (26/26)	24.3 s (L4, 1 at a time) / 3.3 s (H100, 1 at a time)	29.1 s (L4, 1 at a time) / 3.8 s (H100, 1 at a time)	$0.47 (L4, 16 concurrent) / $5.36 (L4, 1 at a time) / $3.59 (H100, 1 at a time)
Qwen3.5-0.8B (best prompt)	66.0	97.0	91.7	95.3	93.3	87.7	6.9 (2/29)	7.7 (2/26)	8.7 s (L4, 16 concurrent)	10.9 s (L4, 16 concurrent)	$0.12 (L4, 16 concurrent)
Qwen3.5-2B (best prompt)	68.3	97.7	91.7	94.0	94.3	88.0	17.2 (5/29)	34.6 (9/26)	14.1 s (L4, 16 concurrent)	19.5 s (L4, 16 concurrent)	$0.20 (L4, 16 concurrent)
Qwen3.5-4B (best prompt)	73.3	68.7	94.3	91.0	90.0	90.7	82.8 (24/29)	96.2 (25/26)	31.1 s (L4, 16 concurrent)	40.0 s (L4, 16 concurrent)	$0.44 (L4, 16 concurrent)
GPT-6 Astra (best prompt)	99.0	99.7	100.0	99.3	99.7	100.0	100.0 (29/29)	100.0 (26/26)	12.1 s (API)	15.3 s (API)	$175.31 (API)
Claude Opus 5.5 (best prompt)	97.7	100.0	100.0	98.7	99.0	99.7	100.0 (29/29)	100.0 (26/26)	11.5 s (API)	16.8 s (API)	$116.41 (API)
Gemini 4 Argon (best prompt)	98.0	100.0	100.0	98.7	99.0	99.7	100.0 (29/29)	100.0 (26/26)	8.6 s (API)	32.7 s (API)	$36.92 (API)

Table 2. Gain from each pipeline stage (final model's pipeline, fresh eval_id v2, 300 cases)

Stage	Decision acc %	Δ decision (pts) [95% CI]	Risk exact %	Δ risk	Actions recall %	Δ actions	Screening macro-F1 %	Δ screening	Unsupported claims %	Δ unsupported
Qwen3.5-4B, zero-shot	56.3	–	34.0	–	33.1	–	23.6	–	59.4	–
+ best prompt (rulebook in the prompt)	56.0	-0.3 [-6.0, +5.7]	50.3	+16.3	34.6	+1.5	34.6	+11.0	52.0	-7.4
+ domain adapter + Counterfactual Twin Distillation (short prompt, no pre-checks)	88.7	+32.7 [+26.0, +39.3]	88.3	+38.0	85.2	+50.6	97.4	+62.8	16.2	-35.8
+ code pre-checker (reader v3, trained with 25% blank blocks)	96.7	+8.0 [+4.3, +11.7]	96.0	+7.7	95.1	+9.9	99.5	+2.1	13.7	-2.5
+ post-train: SFT on exact labels (= final model, B2)	96.3	-0.3 [-1.7, +0.7]	95.3	-0.7	93.6	-1.5	99.6	+0.1	13.7	-0.0

Appendix A. A, B and B2 against the frontier, every test set (decision accuracy %)

A = model alone. B = model + code pre-checker (reader v2), the result before the diagnosis. B2 = final model: pre-checker reader v3, trained with 25% blank blocks, after the diagnosis.

Model	eval_id (300)	OOD template (50)	OOD hand (50)	OOD LLM (100)	Frozen 500	Hard (150)	Fresh eval_id v2 (300)
A: model alone	88.3	84.0	82.0	94.0	88.4	85.3	–
B: model + pre-checker v2 (before diagnosis)	93.0	90.0	88.0	86.0	90.8	93.3	–
B2: model + pre-checker v3, blank-block training (shipped model)	96.0	94.0	92.0	92.0	94.6	92.7	96.3
GPT-6 Astra (best prompt)	99.0	100.0	100.0	99.0	99.2	99.3	99.3
Claude Opus 5.5 (best prompt)	98.0	100.0	100.0	98.0	98.4	98.7	97.7
Gemini 4 Argon (best prompt)	94.0	100.0	98.0	98.0	95.8	–	95.0

Appendix B. Initial benchmark: open and frontier models, zero-shot and best prompt (frozen eval v1, 500 cases)

Model	Prompt	Params	Decision acc %	Risk exact %	Actions recall %	Screening macro-F1 %	Unsupported claims % (LLM judge)	Schema valid %	$ / 1k cases
GPT-6 Astra	zero-shot	undisclosed	92.6	73.4	96.0	84.0	18.2	100.0	$108.29
GPT-6 Astra	best prompt	undisclosed	99.2	99.2	99.4	100.0	7.3	100.0	$176.17
Claude Opus 5.5	zero-shot	undisclosed	87.6	77.6	93.9	89.3	18.5	100.0	$75.01
Claude Opus 5.5	best prompt	undisclosed	98.4	96.0	98.1	100.0	8.3	100.0	$116.67
Gemini 4 Argon	zero-shot	undisclosed	91.4	81.2	89.4	98.1	22.7	100.0	$22.24
Gemini 4 Argon	best prompt	undisclosed	95.8	96.0	94.4	100.0	8.8	100.0	$37.05
Qwen3.5-0.8B	zero-shot	0.8B	40.0	24.0	25.5	6.9	73.7	100.0	$0.14
Qwen3.5-0.8B	best prompt	0.8B	39.6	21.6	3.7	21.4	63.6	100.0	$0.12
Qwen3.5-2B	zero-shot	2B	29.0	29.2	49.1	38.2	64.2	99.6	$0.26
Qwen3.5-2B	best prompt	2B	39.4	36.2	29.4	44.3	45.9	100.0	$0.20
Qwen3.5-4B	zero-shot	4B	60.0	34.8	38.5	25.8	59.7	100.0	$0.36
Qwen3.5-4B	best prompt	4B	61.2	51.6	37.2	34.7	53.1	100.0	$0.44

Appendix C. Model size and distillation ablation (dev, 150 cases; model selection only, never the test sets)

Rejection sampling scored one case higher than Counterfactual Twin Distillation on decision accuracy (a tie within noise); twins was chosen after seeing dev for its required-actions recall and fewest wrong approvals, and that choice is stated as post-hoc.

Run (dev, 150 cases)	Decision acc %	Risk exact %	Field acc % (mean)	Actions recall %	Screening macro-F1 %
Qwen3.5-0.8B, trained on the answer key	84.0	84.7	97.6	79.5	100.0
Qwen3.5-0.8B, answer key + Counterfactual Twins	87.3	84.7	98.1	83.7	100.0
Qwen3.5-2B, trained on the answer key	86.7	80.7	97.6	79.5	100.0
Qwen3.5-4B, trained on the answer key (ceiling / control)	90.0	89.3	97.7	85.3	100.0
4B distillation: floor (Sol answers)	89.3	88.0	98.4	88.4	100.0
4B distillation: + rejection sampling (code checks)	91.3	90.7	98.1	84.2	97.2
4B distillation: + multi-teacher arbitration (Sol / Argon)	90.0	89.3	98.1	86.3	100.0
4B distillation: + Counterfactual Twin Distillation (chosen)	90.7	90.7	98.8	90.5	100.0
4B: twins on the domain-adapted base	91.3	90.7	98.7	90.5	100.0